

Pregnancy Success Probability Prediction Using Infertility Treatment Data

LG AI Hackathon

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* CONTENTS

1. PROBLEM STATEMENT & DATASET OVERVIEW
2. EDA & PROCESSING
3. MODELING
4. INSIGHTS
5. PRACTICALITY AND APPLICABILITY OF THE FINAL MODEL
6. EXPERIMENTS

* PROBLEM STATEMENT & DATASET OVERVIEW

Goal & Background

To reduce the treatment burden on patients undergoing infertility procedures and enable medical institutions to offer differentiated services, we **develop an AI model that predicts the probability of pregnancy success** using infertility patient data and explores the optimal features influencing pregnancy outcomes.

Data Introduction

- Train Data: Infertility treatment data including pregnancy success labels with a unique sample ID, containing 126,244 records and 34 columns.
- Test Data: Infertility treatment data with unique sample IDs, containing 54,412 records and 33 columns.

'Age at Procedure', 'Treatment Type', 'Total Number of Pregnancies', 'Number of Embryos Transferred', 'Source of Sperm'

환자 시술 당시 나이, 시술 유형, 이전 총 임신 성공 횟수, 이식된 배아 수, 정자 출처 등..

* EDA & PREPROCESSING

Variable Type Conversion

'Age at Procedure', 'Donor Oocyte Age', 'Donor Sperm Age' 환자 시술 당시 나이, 난자 기증자 나이, 정자 기증자 나이

Converted categorical variables to numeric by mapping each range to its median value, and assigned -1 to “Unknown” values for identification.

Columns related to number of oocytes and embryos 난자 수, 배아 수 관련 칼럼

Converted range values to their numeric maximum (e.g., '1-5' → 5).

Merged the last two ranges when they showed similar pregnancy success rates and contained small amounts of data, as the separation was deemed insignificant.

Columns related to counts 횟수 관련 칼럼

Processed variables as numeric since their meaning represents numerical quantities.

* EDA & PREPROCESSING

Missing Value Handling

'Total Number of Pregnancies' 이전 총 임신 횟수

If 'Total Number of Pregnancies' is missing but 'Total Number of Births' is not, replace the missing value with the number of births, assuming the patient was pregnant at least that many times.

이전 총 임신 횟수 결측	이전 총 성공 임신 횟수 결측	count
False	False	114541
True	False	6622
True	True	5081

Other Missing Values

Replace all remaining missing values with -1.

* EDA & PREPROCESSING

Variable Removal

'Fresh Oocyte Use' 신선 난자 사용 여부

Found to have a distribution inconsistent with the related variable 'Number of Freshly Retrieved Oocytes'.
⇒ Retained 'Number of Freshly Retrieved Oocytes', which contains more information,
and removed 'Fresh Oocyte Use'.

신선 난자 사용 여부	채취된 신선 난자 수	개수
0	0	39876
0	1	85963
1	1	405

* EDA & PREPROCESSING

'Specific Treatment Type'	count	mean_success_rate
세부 시술 유형		
ICSI / AH:Unknown	1	1.000000
ICSI / BLASTOCYST:ICSI	1	1.000000
IVF / BLASTOCYST	1062	0.349509
ICSI / BLASTOCYST	1356	0.348777
FER	3	0.333333
IVF:ICSI	350	0.309524
ICSI / BLASTOCYST:IVF / BLASTOCYST	10	0.300000
IVF	43912	0.270535
ICSI	56714	0.269160
Unknown	18380	0.257319
ICSI:IVF	863	0.252646
ICSI / AH	684	0.209969
IVF / AH	303	0.202420
IVF:Unknown	96	0.187500
ICSI:Unknown	199	0.132328
DI	847	0.124224
ICSI:ICSI	910	0.025275
IVF:IVF	550	0.021818
IVF / AH:ICSI / AH	1	0.000000

Unique Values in Categorical Variable

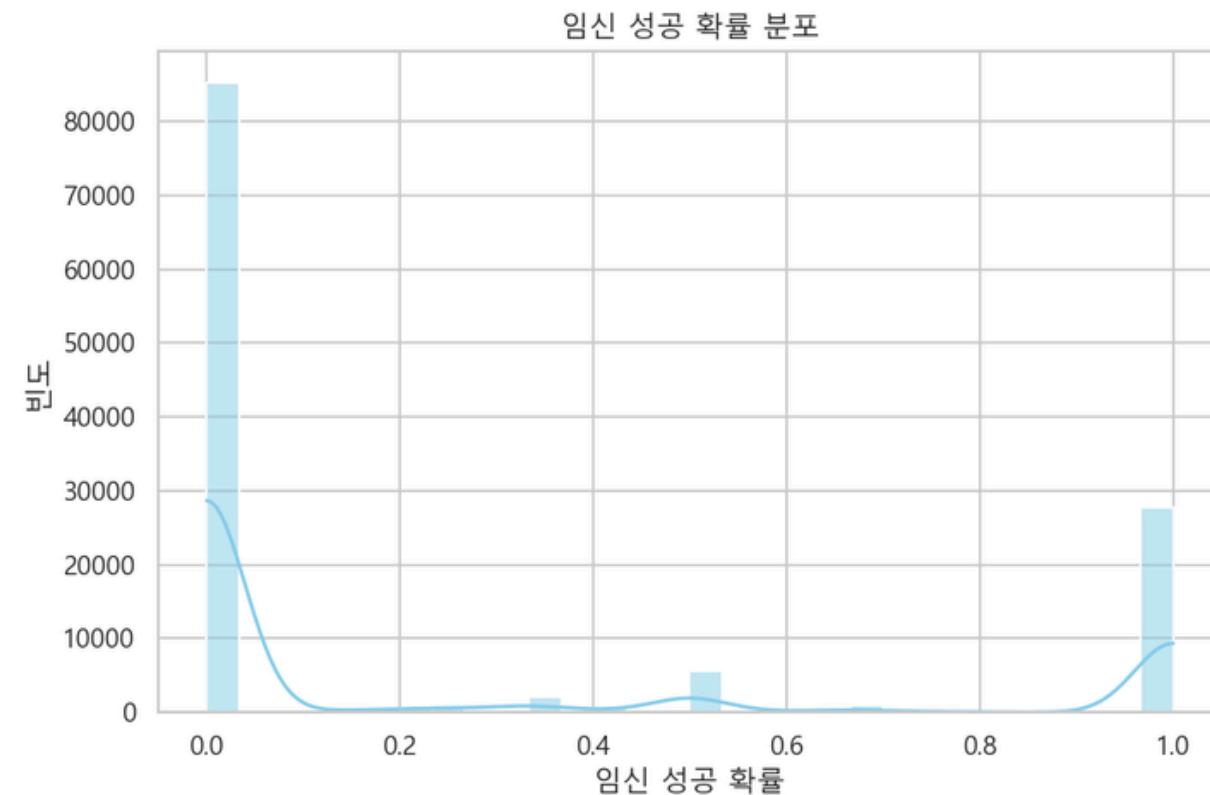
Cases containing '**BLASTOCYST**' tend to have a **higher** average pregnancy success rate overall.

1. When 'BLASTOCYST' is included → 'BLASTOCYST'
2. When 'AH' is included → 'AH'
3. When the same treatment type is connected with “:” → 'DOUBLE'
4. When different treatment types are connected with “:” → 'DOUBLE_CROSS'
5. When 'Unknown' is included with another treatment → 'DOUBLE_UNKNOWN'
6. 'ICSI', 'IVF', 'Unknown' remain unchanged
7. All others 'ETC' ⇒ Number of unique values reduced from 19 to 10.

Cases where the same treatment type is connected with “:” tend to have a **lower** average pregnancy success rate overall.

* EDA & PREPROCESSING

Target Distribution



임신 성공 확률	
(0.0, 0.1]	178
(0.1, 0.2]	1178
(0.2, 0.3]	1468
(0.3, 0.4]	2724
(0.4, 0.5]	5911
(0.5, 0.6]	376
(0.6, 0.7]	990
(0.7, 0.8]	356
(0.8, 0.9]	46
(0.9, 1.0]	27794

- The target variable (pregnancy success probability) in the final round is a continuous value between 0 and 1 (regression task).
- Approximately 90% of the values are either 0 or 1, while the remaining 10% are distributed continuously across intermediate values.

* MODELING

Problem Definition

Binary classification model

- Validation set construction reflecting **target imbalance**
- Output extracted as **probability values** within valid range

Regression model

- Difficulty in reflecting target ratio when constructing validation set
- Need to clip output probabilities within valid range



* MODELING

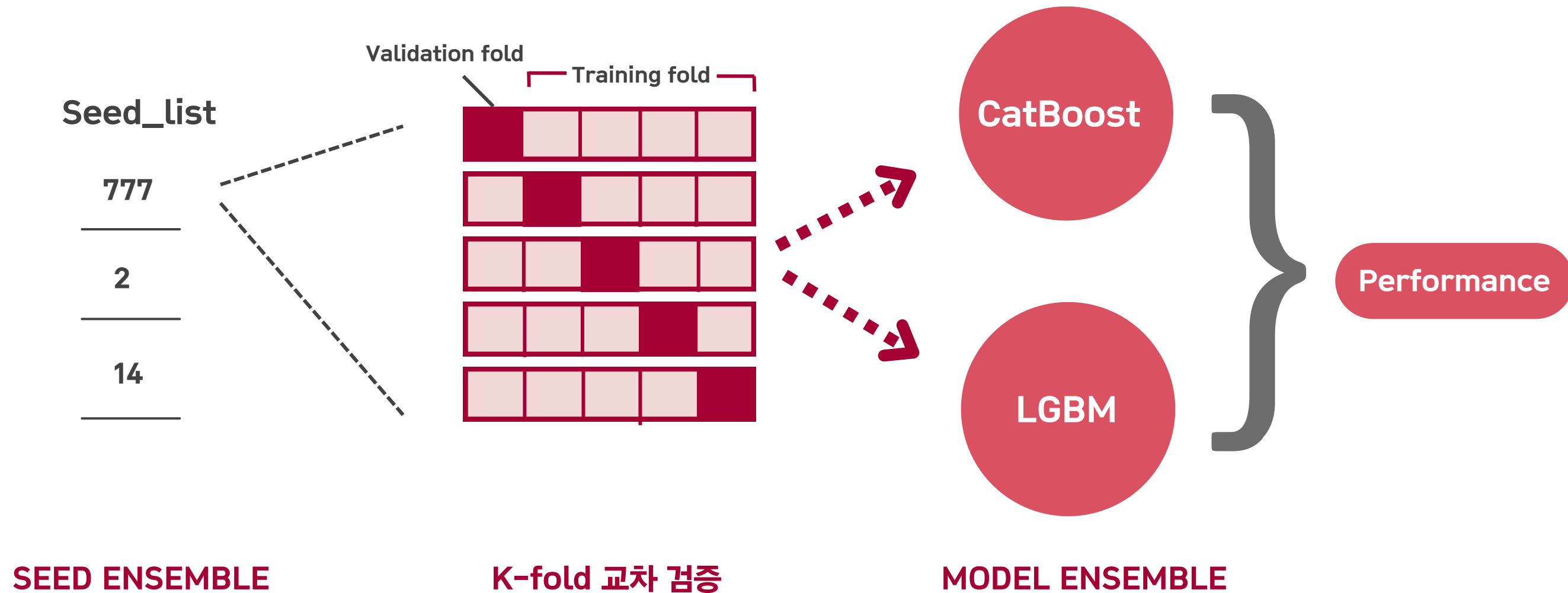
StratifiedKFold + class weight

Model training and experiments were conducted using K-fold cross-validation.

To address target imbalance, data was split into folds with identical target distributions using **Stratified K-fold**. 5-fold cross-validation was used to generalize model performance and prevent overfitting.

Class weights were applied to the model using the target ratio **within the train data of each fold**. **Target imbalance was reflected in model training**.

* MODELING _SEED ENSEMBLE & MODEL ENSEMBLE



- To prevent overfitting by increasing model generalization and stability, **Seed Ensemble and Model Ensemble** were used together.
- **CatBoost**: Handles categorical variables using a sequential statistical encoding method.
- **LGBM**: Learns diverse data characteristics through flexible splitting using an asymmetric tree structure.
- Both models were ensembled to leverage the strengths of each.

* MODELING

Hyperparameter Tuning

Used Optuna to tune LightGBM and CatBoost hyperparameters.

⇒ By narrowing the search space considering parameters such as number of trees, learning rate, maximum depth, sampling ratio, L1/L2 regularization, and gamma, the optimal hyperparameter set was obtained and applied to the models.

* INSIGHTS

Class Weight Experiments

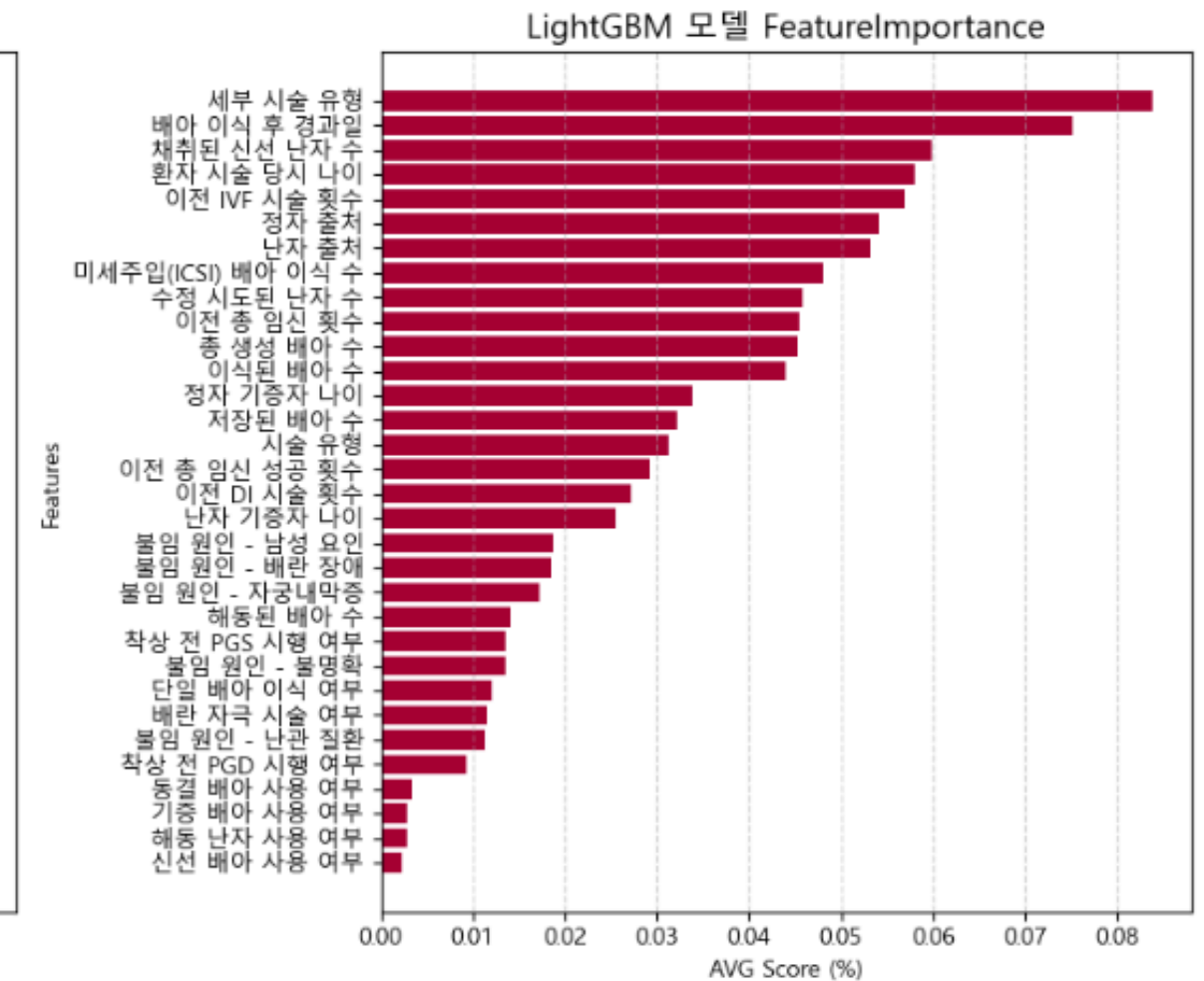
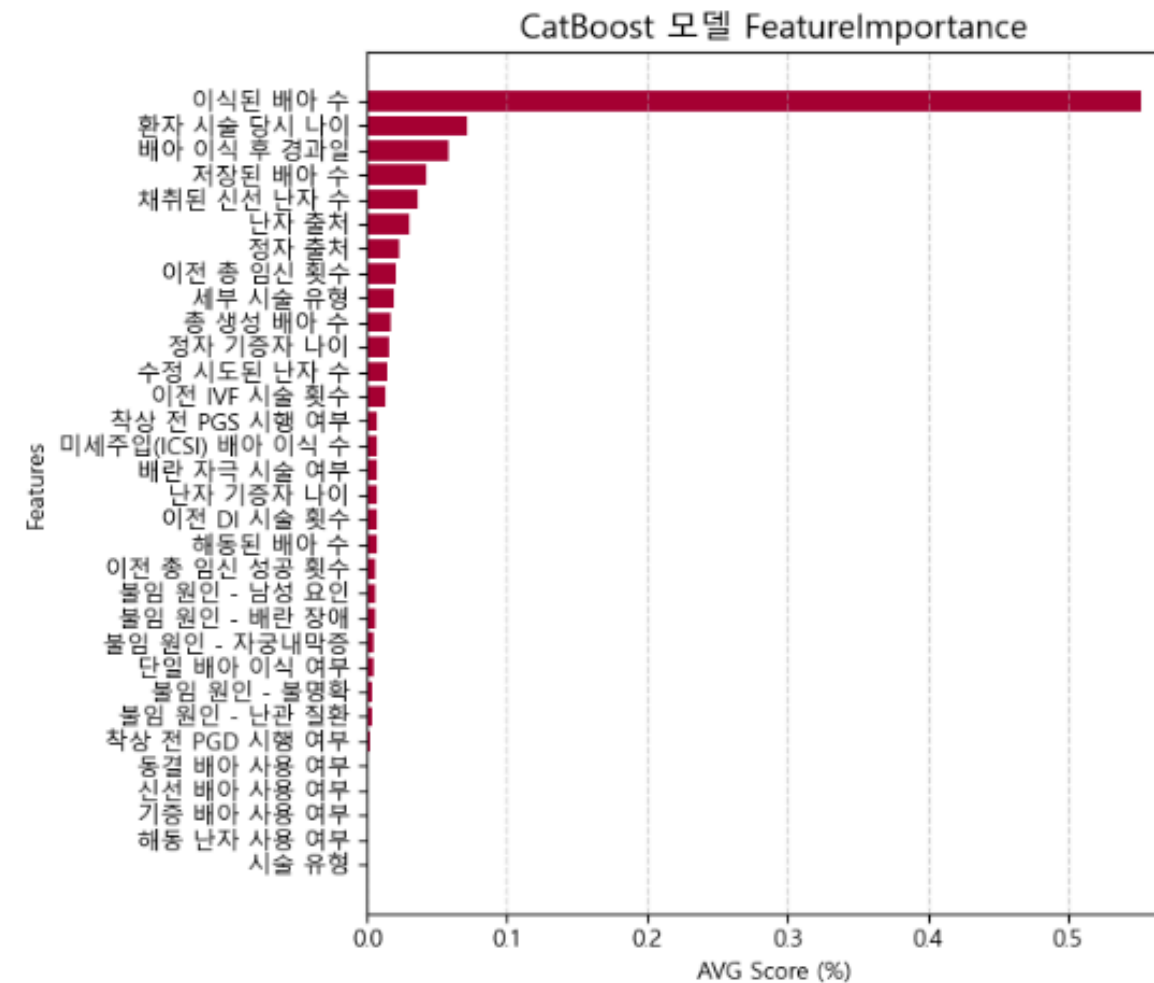
	Balanced=False	Balanced=True
XGBoost	0.4203	0.6479
LightGBM	0.4211	0.6579
CatBoost	0.4308	0.6581

- To evaluate the effect of Class Weight, experiments were conducted by averaging performance over 3 seeds and 5-folds per model.
- All three models showed **performance improvements exceeding 20 percentage points when Class Weight was applied during training.**

* INSIGHTS

Feature Importance

CatBoost(Symmetric)에서는 '이식된 배아 수' 등 모든 시술에서 중요한 변수들의 중요도가 높게 학습
LightGBM(Asymmetric)에서는 '세부 시술 유형'의 중요도가 높아 시술 유형에 따라 다양한 Rule을 학습



- In **CatBoost (Symmetric)**, important variables across all procedures—such as ‘Number of Embryos Transferred’—were learned with high importance.
- In **LightGBM (Asymmetric)**, the variable ‘Specific Treatment Type’ showed high importance, **reflecting that the model learned diverse rules depending on treatment types.**

* Practicality and Applicability of the Final Model

```
=====
Seed Number : 777
=====
<Model : LGBM>
[Valid] Fold #1 Score: 0.6638
[Valid] Fold #2 Score: 0.6577
[Valid] Fold #3 Score: 0.6551
[Valid] Fold #4 Score: 0.6552
[Valid] Fold #5 Score: 0.6576
Avg. Score of validset: 0.6578974152274235
Std. Score of validset: 0.0031739081839709158

<Model : CB>
[Valid] Fold #1 Score: 0.6619
[Valid] Fold #2 Score: 0.6577
[Valid] Fold #3 Score: 0.6544
[Valid] Fold #4 Score: 0.6559
[Valid] Fold #5 Score: 0.6582
Avg. Score of validset: 0.657637377928958
Std. Score of validset: 0.0025436854621755265

Seed Number : 2
=====
<Model : LGBM>
[Valid] Fold #1 Score: 0.6564
[Valid] Fold #2 Score: 0.6549
[Valid] Fold #3 Score: 0.6560
[Valid] Fold #4 Score: 0.6580
[Valid] Fold #5 Score: 0.6597
Avg. Score of validset: 0.6570215218873903
Std. Score of validset: 0.0016813447877224383

<Model : CB>
[Valid] Fold #1 Score: 0.6569
[Valid] Fold #2 Score: 0.6556
[Valid] Fold #3 Score: 0.6563
[Valid] Fold #4 Score: 0.6580
[Valid] Fold #5 Score: 0.6592
Avg. Score of validset: 0.6572010668291671
Std. Score of validset: 0.0012703523155506793

Seed Number : 1234
=====
<Model : LGBM>
[Valid] Fold #1 Score: 0.6571
[Valid] Fold #2 Score: 0.6581
[Valid] Fold #3 Score: 0.6548
[Valid] Fold #4 Score: 0.6593
[Valid] Fold #5 Score: 0.6565
Avg. Score of validset: 0.6571594308876059
Std. Score of validset: 0.0015107519394656212

<Model : CB>
[Valid] Fold #1 Score: 0.6570
[Valid] Fold #2 Score: 0.6576
[Valid] Fold #3 Score: 0.6558
[Valid] Fold #4 Score: 0.6599
[Valid] Fold #5 Score: 0.6576
Avg. Score of validset: 0.6575737545677491
Std. Score of validset: 0.0013409329053613481

총 실행 시간: 8분 19.28초
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After various experiments combining model ensembles, hyperparameter tuning, seed ensembling, and SMOTE-based oversampling,

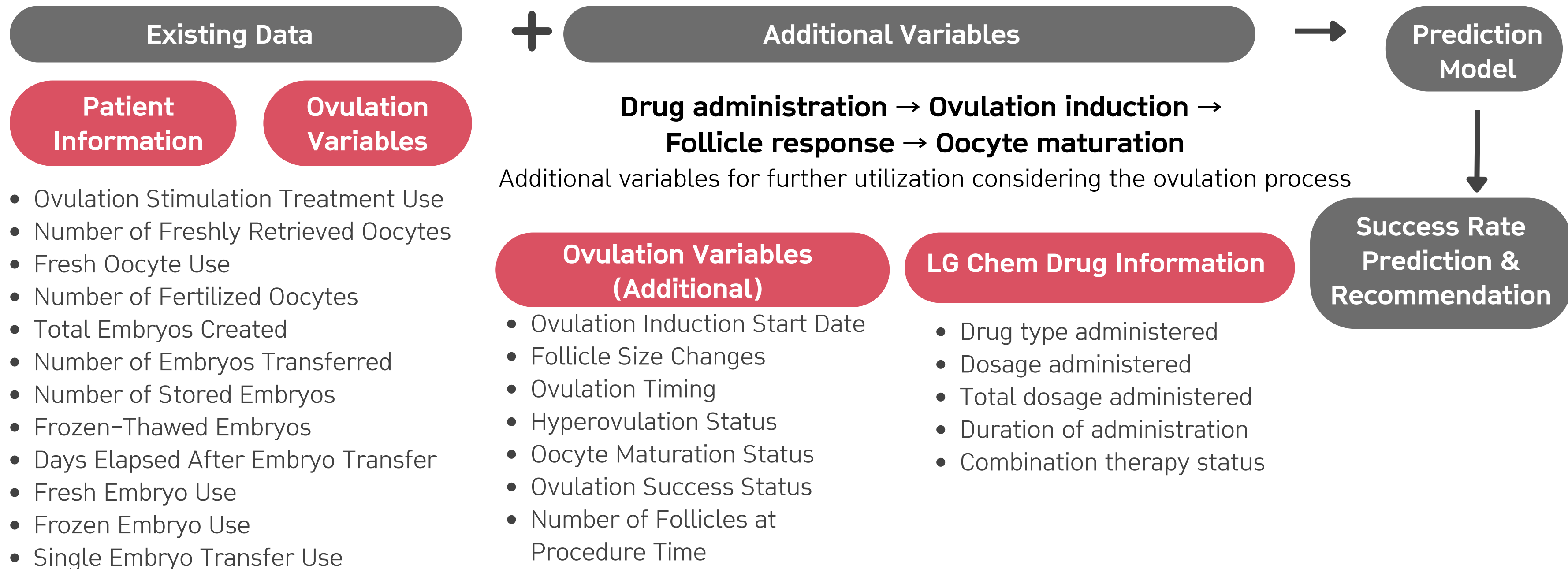
the final optimal model was selected as a **3-seed ensemble of LightGBM and CatBoost models** with **Optuna-based hyperparameter optimization** applied.

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3	엄마손은 AI손	    	0.6648	42

- **Speed:** Reduced training time (8 minutes) and inference time (1 second) → Enables monthly batch training and real-time inference.
- **Generalization Performance:** High private score → Ensures stable performance in real-world environments.

* Practicality and Applicability of the Final Model

💡 Utilizing Existing Models & Data When Adding New Variables



* Practicality and Applicability of the Final Model

💡 Expected Benefits

Medical staff

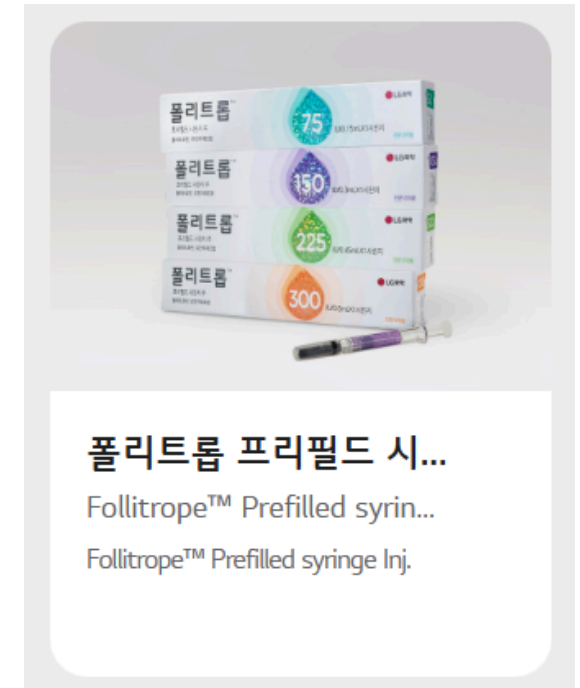
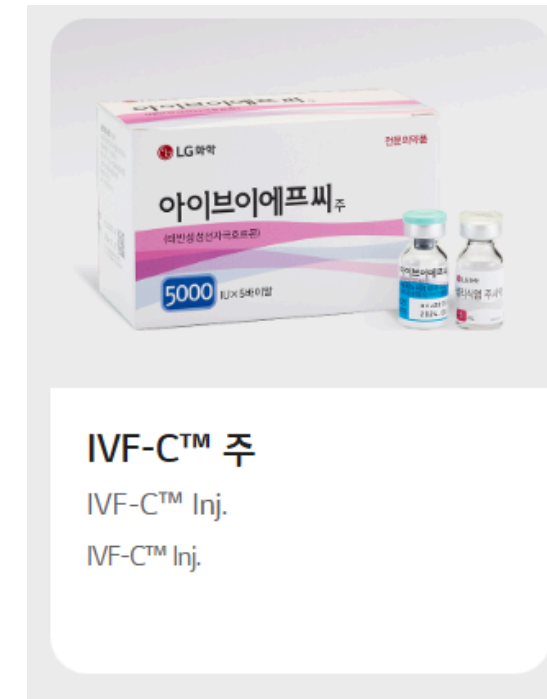
- Enables personalized treatment planning
- Improves procedure success rates
- Supports clinical decision-making

Patients

- Recommends procedures with higher chances of success
- Reduces patient burden from repeated treatments

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- Demonstrates product efficacy through data
- Secures persuasive value for clients (hospitals)
- Provides differentiated solutions compared to competitors



* EXPERIMENTS

Derived Variables & Preprocessing Experiments

Variables that were tested but **did not contribute** to performance improvement

① Infertility cause-related variables

(Tubal Disease, Ovulation Disorder, Male Factor, Endometriosis, Unclear Cause)

- Purpose

Due to the presence of multiple binary columns, the number of splits in tree-based models could increase excessively, potentially degrading performance relative to complexity

→ To make the structure simpler without losing data

- Attempts

Various preprocessing methods such as gender-based separation, summation of multiple causes, and variable integration were tried

- Reason for failure

Infertility causes mainly influence the choice of treatment method (not direct causation) + low variable importance

⇒ Judged to have low direct explanatory power, proceeded without additional preprocessing or experiments

* EXPERIMENTS

Derived Variables & Preprocessing Experiments

Variables that were tested but did not contribute to performance improvement

② Embryo-related variables

(Total Embryos Created / Number of Embryos Transferred / Number of Stored Embryos / Frozen-Thawed Embryos)

- Purpose

High variable importance + The embryo transfer process is performed immediately before pregnancy success, considered a key factor with direct impact

→ Expected performance improvement by creating derived variables linked with such numeric variables

- Attempts

Created derived variables such as fresh embryo ratio, embryo transfer status, and embryo creation success rate

- Reason for failure

Possible redundancy of information among variables + risk of overfitting

$$\frac{\text{Total Embryos Created}}{\text{Number of Embryos Transferred} + \text{Number of Stored Embryos}}$$

$$\frac{\text{Total Embryos Created}}{\text{Number of Fertilized Oocytes}}$$

* EXPERIMENTS

Derived Variables & Preprocessing Experiments

Variables that were tested but did not contribute to performance improvement

③ Procedure/Pregnancy Count Scaling

(Previous IVF/DI Treatment Count, Previous Total Number of Pregnancies, Previous Total Number of Births)

- Purpose

Procedure and pregnancy counts require relative evaluation depending on pregnancy attempt duration or patient age
→ Attempted scaling procedure/pregnancy count variables by patient age

- Attempts

Created derived variables such as treatment count per age, pregnancy count per age, and birth count per age

- Reason for failure

Low variable importance + failure to consider changes in treatment type, fertile period, and handling of zero values

THANK YOU

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고경준 김예진 나영은 이수민 정서윤